

SOHAN DAS

ROBOTICS ENGINEER

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SUMMARY

I am an MSc Robotics student at the university of Manchester with good academic and technical foundation in control systems, sensor fusion and embedded electronics besides core robotics concepts. I have experience in both research and hands-on robotics projects to designing custom PCBs for real-world applications. Passionate about applying advanced robotics to challenging real-world domains including space robotics, in-orbit servicing, assembly and manufacturing (ISAM). Self-motivated, detail-oriented, and eager to contribute to innovative UK space robotics capability.

SKILLS

- MATLAB / Simulink
- Linux
- Git / Github
- Arduino and Raspberry Pi
- ROS2
- Nav2 / SLAM / Robot Kinematics
- Gazebo / RViz
- MuJoCo, FANUC, Cyberbotics, RoboDK
- Python
- HTML, CSS and JavaScript
- PCB Design/Control System/PID
- SQL
- R
- Embedded C
- Java / C++
- Computer Vision and Cognitive Robotics

LANGUAGE

- English
- Bengali (Native)
- Hindi (Fluent)
- Japanese (Basic)

PUBLICATION

Arm Motion Planning of Humanoid Robot
(DOI:10.13140/RG.2.2.33678.54082)

LINKS

 [CPD Portfolio](#)

 [Github](#)

 [Linkedin](#)

EXPERIENCE

Virtual Robotics and Control Engineer

Forage

Feb, 2026 - Mar, 2026

- Completed hands-on coding tasks simulating real-world engineering workflows
- Implemented solutions using structured problem-solving and debugging techniques
- Applied programming concepts to solve industry-relevant scenarios
- Strengthened understanding of software development practices and code optimization

RPA Developer

Automation Anywhere

Jun, 2022 - Jul, 2022

- Collaborated with engineering teams to develop robotic process automation (RPA) workflows to streamline repetitive enterprise tasks
- Gained practical experience with the Automation Anywhere platform and bot deployment lifecycle

NOTABLE PROJECTS

- **Leo-Rover robot:** Developed an autonomous Leo Rover platform on ROS 2 Jazzy capable of mapping its environment, navigating to goal locations, and integrating with vision-based object detection and a manipulator arm for pick-and-place operations.
- **BTech Thesis:** Comparative analysis of DDPG, SAC and TD3 deep RL algorithms for vibration suppression on Quanser SRV02 physical rotary flexible link hardware — MATLAB/Simulink + RL Toolbox; SAC achieved superior sample efficiency and robustness
- **Autonomous Robot Navigation using ROS 2 Navigation Stack:** Full Nav2 pipeline on Leo-Rover: SLAM, localization, path execution — design through integration testing in Linux/ROS2
- **Real-Time Autonomous Control System for Aerial Object Navigation**
- **Path Planning (A* Algorithm)**
- **UAV Position Controller:** Cascade PID + Disturbance Observer for real-time aerial stabilization; prototyped in MATLAB/Simulink, implemented in embedded C
- **Electric Vehicle Battery Management System:** Safety-conscious embedded firmware with I2C/SPI peripheral comms, cell balancing, and protection cutoffs; PCB designed in OrCAD PSpice

EDUCATION

- **Master of Science in Robotics** Sept, 2025 - Present
University of Manchester
- **Bachelor of Technology in Electrical and Electronics Engineering** Aug, 2020 - Jul, 2024
VIT UNIVERSITY
GPA: 8.46 / 10.0